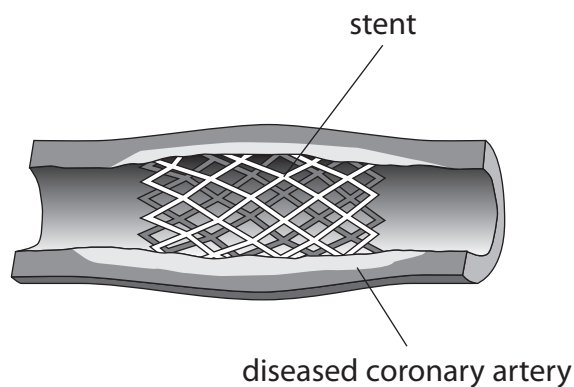


5 Stents are used in the treatment of atherosclerosis.

Stents widen the diseased coronary artery so that blood can flow through to the heart muscle.

The diagram shows a stent.



Stents can damage the endothelial cells lining the artery and trigger the formation of a blood clot.

Patients who have a stent fitted are given anticoagulants.

(a) Explain how damage to the endothelial cells results in the formation of thrombin.

(2)

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(b) Give **one** risk of treating patients with anticoagulants.

(1)

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(c) The photograph shows a tropical bont tick.



(Source: © Phanie/Alamy Stock Photo)

This bont tick produces a peptide called variegain.

Variegain binds to the active site of thrombin and prevents blood clotting.

(i) Variegain is a peptide consisting of 32 amino acids.

Which is the minimum number of bases in the gene that codes for this peptide?

(1)

- ☐ A 32
- ☐ B 33
- ☐ C 96
- ☐ D 99

(ii) Explain how variegain prevents blood clotting.

(2)

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(iii) Thrombin slowly splits variegain into two molecules.

Explain how thrombin splits variegain.

(2)

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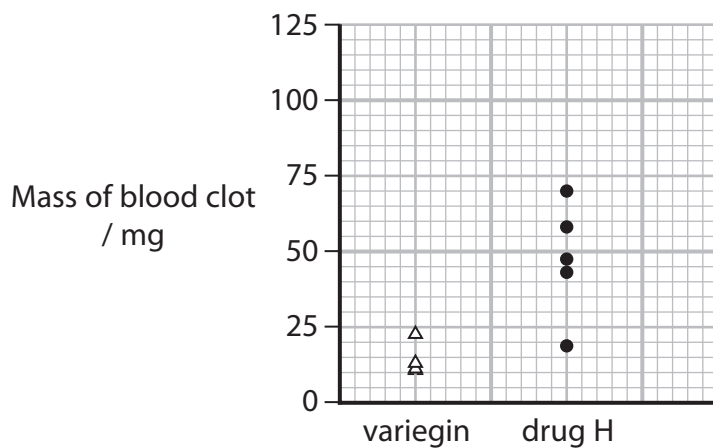


- (iv) The effect of variegatin on the mass of blood clots forming inside a stent was investigated.

The masses were compared with another anticoagulant, drug H.

The graph shows the results of this investigation.

Each plotted point represents the mass of blood clot in one stent.



Comment on the results of this investigation.

(2)

(Total for Question 5 = 10 marks)